

Anaïs Sallard

HARDWARE & EMBEDDED SYSTEMS ENGINEER

PROFILE

Hardware & Embedded Systems Engineer with **two years of experience in critical space systems.** Hands-on experience in **digital electronics, FPGA/VHDL, electronic system modelling, sensor interfaces and hardware/software integration**. Strong background in **low-level embedded development and real-time systems**, with additional experience in AI and radar applications.

CONTACT

Singapour

anais.sallard@edu.esiee.fr

TECHNICAL SKILLS

Hardware & Digital Design
FPGA • VHDL • Digital Electronics • FSM • RTL Design • Sensor Interfaces

Embedded Systems
Embedded C/C++ • Real-Time Systems
• **Low-Level Drivers**
• **Hardware/Software Integration**
• Interrupts • Memory-Mapped I/O

Interfaces & Electronics
UART • SPI • SpaceWire • PSI5
• Analog Electronics • RFID
• Signal Acquisition

Hardware Tools & Simulation
Xilinx Vivado • QuestaSim
• Cadence PSpice
• MATLAB/Simulink • LabVIEW

Software & Development Tools
Python • Git • Jenkins • CMake
• Google Test • Linux

AI - Additional Expertise
Machine Learning • Deep Learning
• PyTorch/TensorFlow
• Reliability & Latent-Space Analysis

PROFESSIONAL EXPERIENCE

THALES SOLUTIONS ASIA — AI Engineering Intern
Singapore | Aug 2026 - Nov 2026

- Developed a model reliability assessment approach based on the analysis of latent-space representations.
- Analysed the relationship between latent-space structure and model prediction reliability to identify less trustworthy predictions.

SODERN (ArianeGroup) — Embedded Software Engineer, Apprenticeship
Paris, France | 2024 - 2026

- Developed **low-level embedded C software** for the **GR740 LEON4 SPARC** space processor.
- Designed and validated a real-time **UART driver**, including RX/TX interrupts, FIFO management, callbacks and timeout handling.
- Integrated a **SpaceWire service into the XtratuM hypervisor** environment.
- Performed l**ow-level debugging and hardware/software integration** using **hardware registers, interrupts and memory-mapped interfaces**.
- Conducted embedded software verification and validation on real-time systems.

SELECTED HARDWARE & EMBEDDED PROJECTS

13.56 MHz HF RFID System — RF & Analog Electronics
• **Modelled and simulated** an **ISO14443** 13.56 MHz **RF reader/tag system**, including antenna resonance, **magnetic coupling** and impedance matching.
• **Analysed RF power transfer**, bandwidth and load modulation for passive tag communication using Cadence PSpice.

RF Antenna Design & Propagation — 2.45 GHz

- **Modelled and simulated dipole, monopole and directional antennas** at 2.45 GHz using 4NEC2/NEC2.
- **Analysed impedance matching, SWR, bandwidth, gain and radiation patterns**, as well as **RF propagation**, received power and antenna **polarization**.

FPGA Digital System Design — VHDL / Xilinx

- Designed and implemented FPGA-based digital systems in VHDL, including datapath and finite-state machine (FSM) architectures.
- Developed SPI sensor interfacing and UART communication, with **simulation and hardware validation** on Basys 3 FPGA boards.
- Performed functional **verification** using QuestaSim and Xilinx Vivado.

PSI5 Sensor Interface — Mixed-Signal Electronics

- Analysed and simulated the **physical layer** of a PSI5 sensor interface for bidirectional ECU-**sensor** communication.
- **Studied voltage/current modulation**, analog interface circuits and Manchester-coded communication using **Cadence PSpice.**

EDUCATION

2023–2026 - ESIEE Paris
Engineering Degree in Embedded Systems (Apprenticeship Program)
Digital Electronics, FPGA/VHDL, Embedded C Programming, Real-Time Systems

2021–2022 - Preparatory Classes for Engineering Schools (CPGE)
Physics-Chemistry (PC) - Lycée Lavoisier, Paris, France
Physics, Chemistry and Engineering Sciences (PCSI) - Lycée Raspail, Paris, France